

Introduction to Database Management System	
BVNSD 1.4	
Credits:4	40 Hours
Introduction:	
The DBMS (Database management system) is one of the most important module because this module it is learn how the records are maintain, how the records are store and retrieve. File management concept and record managements concept are also cover in this module.	
Course Objectives:	
The main objective of this module is to give fundamental knowledge of data base, data base management.	
Note: 3 Credits for Theory and 1Credit for Practical	
Unit 1:	7 Hours
An overview of database management system, Advantages of Database Management System, Database System vs. File System, Database System Architecture, Data Models Schema and Instances, Data Independence and database Language and Interfaces, Data Definitions Language,DML, Overall Database Structure.	
Unit 2:	7 Hours
ER model concepts, notation for ER diagram, mapping constraints, keys, Concepts of Super Key,candidate key, primary key, Generalization, aggregation, reduction of an ER diagrams to tables, extended ER model, relationships of higher degree.	
Unit 3:	7 Hours
Relational data model concepts, integrity constraints: entity integrity, referential integrity, Keys constraints, Domain constraints, relational algebra, relational calculus, tuple and domain calculus.	
Unit 4:	7 Hours
Characteristics of SQL; Advantages of SQL; SQL Data Types and Literals; DDL,DML, DCL, and TCL; Controlling user access; SQL Operators and their Procedure; Adding/ Alerting/ Dropping Constraints; Views and Indexes; Queries and sub queries/ nested queries; Aggregate Functions, Insert (single & bulk), Update and Delete Operations in SQL; Different types of joining; Date Functions; Type Conversion Functions.	
Practical:	12 Hours
Part I: Getting familiar with SQL (Maximum number of turns allotted: 3) 1) Creating tables. 2) Insertion, Deletion, Updation and Retrieval of data. 3) Arithmetic operations, Logical operations and Pattern matching. 4) Concept of Grouping (Group by clause, Having Clause). 5) Use Aggregate function in query. 6) Write commands for Joins, Union and Intersection. 7) Concept of Sub-query. 8) Concept of Data constraints (Unique Key, Primary Key, Foreign Key). 9) Creating Views and Indexes.	
Part II: Relational Database Implementation Implement the following mini-project's database schemas and give an expression in SQL for each of the queries. Project 1. Library Management System: Create the following schema, enter at least 5 records in each table and answer the queries given below. LibraryBooks (Accession number, Title, Author, Department, PurchaseDate, Price) IssuedBooks (Accession number, Borrower)	

- a) Identify primary and foreign keys. Create the tables and insert at least 5 records in each table.
- b) Delete the record of book titled “Database System Concepts”.
- c) Change the Department of the book titled “Discrete Mathematics” to “CSE”.
- d) List all books that belong to “CSE” department.
- e) List all books that belong to “CSE” department and are written by author “Navathe”.
- f) List all computer (Department=“CSE”) that have been issued.
- g) List all books which have a price less than 500 or purchased between “01/01/2015” and “01/01/2019”.

Project 2. Student Management System:

Create the following schema, enter at least 5 records in each table and answer the queries given below.

Student (College roll number, Name of student, Date of birth, Address, Marks(rounded off to whole number) in percentage at 10 + 2, Phone number)

Paper Details (Paper code, Name of the Paper)

Academic_details (College roll number, Paper code, Attendance, Marks in home examination)

- a) Identify primary and foreign keys. Create the tables and insert at least 5 records in each table.
- b) Design a query that will return the records (from the second table) along with the name of student from the first table, related to students who have more than 75% attendance and more than 60% marks in paper 2.
- c) List all students who live in “Lucknow” and have marks greater than 60 in paper 1.
- d) Find the total attendance and total marks obtained by each student.
- e) List the name of student who has got the highest marks in paper 2.

Project 3. Human Resource Management System:

Create the following tables, enter at least 5 records in each table and answer the queries given below.

EMPLOYEE (Person_Name, Street, City)

WORKS (Person_Name, Company_Name, Salary)

COMPANY (Company_Name, City)

MANAGES (Person_Name, Manager_Name)

- a) Identify primary and foreign keys.
- b) Alter table employee, add a column “email” of type varchar(20).
- c) Find the name of all managers who work for both Samba Bank and NCB Bank.
- d) Find the names, street address and cities of residence and salary of all employees who work for “Samba Bank” and earn more than \$10,000.
- e) Find the names of all employees who live in the same city as the company for which they work.
- f) Find the highest salary, lowest salary and average salary paid by each company.
- g) Find the sum of salary and number of employees in each company.
- h) Find the name of the company that pays highest salary.

Course Outcomes(COs):

CO1: Design SQL queries to create database tables and make structural modifications.

CO2: Design SQL queries to add data to the database, edit existing data, and to delete data from the database.

CO3: Implement basic and advanced SQL queries to retrieve data from the database.

CO4: Understand and applies indexing mechanisms in databases.